PERISHABLE GOODS PRICING *Strategy*

INTRODUCTION

Problem	Goal	Key Features
Perishable goods such as fresh food and dairy products face a strict time constraint of their expiration date and if the seller fails to sell all stock in time, there will be a significant direct financial loss	The pricing problem is no longer purely about profit maximization, but also about inventory clearance under time pressure	• Determine the initial optimal price and quantity, considering the holding cost of inventory • Justify the price recommendation that guarantees inventory clearance

METHODOLOGY

1 A linear demand function is estimated using two observed price–quantity data points provided by the user, then a total revenue function is quantity multiplied by demand

2 A total cost function is constructed by combining production cost (variable cost per unit and fixed cost) and inventory holding cost (derived from the Economic Order Quantity (EOQ) model)

3 The initial profit-maximizing price and quantity are the first derivative of the revenue function equal to the first derivative of the cost function

4 After sales begin, inventory is high but the product is approaching its expiration date. A stock-clearing price recommendation is calculated using remaining inventory and remaining time to expiration

DEMAND FUNCTION

We assume a linear demand function:

$$D(p) = k - m.p$$

Where:
- p: selling price per unit
- D(p): quantity demanded per day
- k>0: maximum demand when price is zero
- m>0: sensitivity of demand to price

To estimate k and m, the user inputs two observed price–quantity pairs ***(p1,q1)*** and ***(p2,q2)***

REVENUE FUNCTION

Total revenue function is: $TR = p.q$

Since the product must be sold within T days, total sales is: $q = d.T = (k - m.p)T$

Thus, total revenue becomes:

$$TR = p(k - mp)T = pkT - p^2mT$$

COST FUNCTION

The standard production cost is modeled as: $C(q) = a.q + b$

Where:
- a: cost per unit purchased by the seller
- b: fixed cost

However, for perishable goods, we must also consider holding cost. Let **h: storage cost per unit per day**,

From the EOQ model:
- Average inventory = $\frac{q}{2}$
- Storage duration = $\frac{q}{d}$

Thus, the total holding cost is: $HC = h.\frac{q}{2}.\frac{q}{d} = \frac{hq^2}{2d}$

Hence, the total cost function becomes:

$$TC = C(q) + HC = aq + b + \frac{hq^2}{2d}$$

$$TC = a(k-mp)T + b + \frac{h(k-mp)T^2}{2}$$

INITIAL PROFIT OPTIMIZATION

Thus profit function is:

$$\pi = TR - TC = pkT - p^2mT - a(k-mp)T - b - \frac{h(k-mp)T^2}{2}$$

To maximize profit, we take the derivative of the profit function with respect to price and set it equal to 0:

$$\frac{d\pi}{dp} = 0 \Rightarrow \frac{d}{dp}\left(pkT - p^2mT - a(k-mp)T - b - \frac{h(k-mp)T^2}{2}\right)$$

$$\Rightarrow kT - 2pmt + amT + \frac{hmT^2}{2} = 0$$

So the optimal price is:

$$p_{optimal} = \frac{a}{2} + \frac{k}{2m} + \frac{hT}{4}$$

The corresponding optimal order quantity is:

$$q_{optimal} = (k - mp_{optimal})T$$

STOCK-CLEARING PRICE

On any day during the selling period, when the seller sees that the inventory is still high but the expiration date is approaching. Let:
- $q_{remaining}$: remaining inventory
- $T_{remaining}$: remaining number of selling days
- p_{clear}: stock-clearing price

To fully clear stock, the store must achieve a required daily demand of: $d_{required} = \frac{q_{remaining}}{T_{remaining}}$

Since demand follows the linear function: $d = D(p) = k - mp$

We impose the stock-clearing condition: $d_{required} = k - mp_{clear}$

Thus, stock-pricing price is:

$$p_{clear} = \frac{k - d_{required}}{m} = \frac{k - \frac{q_{remaining}}{T_{remaining}}}{m}$$

WEBSITE-BASED DECISION SYSTEM

Our website implements the entire methodology above as a web-based decision-support tool that:
- Automatically estimates demand functions and cost functions
- Computes optimal price and order size and maximized profit
- Tracks daily sales
- Recommends stock-clearing price

Rather than relying on profit-maximization alone, the stock-clearing approach ensures zero waste, minimized loss, and controlled inventory liquidation.

Gia Huy, Hồng Đăng, Phương Nghi